Firm Characteristics and the Speed of FDA Approval

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This paper empirically examines the effect of firm-specific characteristics on the length of time required by the Food and Drug Administration (FDA) to review and approve new-drug applications between 1990 and 1992. The approach treats regulatory decisions as endogenous and explains the variation in regulatory behavior as a function of differences that exist between firms and drugs. Results show that, controlling for drug-specific characteristics, regulators respond to firm-specific characteristics when evaluating new drug applications. For instance, firms that are less diversified and more R&D-intensive receive shorter review times for their new-drug applications than more diversified and less R&D-intensive firms. The reason is that most firm characteristics signal information to reviewers about either firm reputation or application quality. This information reduces reviewers' uncertainty about approving a dangerous or ineffective drug and leads to faster review times. The results suggest that regulators respond to the heterogeneities among firms in the pharmaceutical market in systematic ways.

1. Introduction

The time required by pharmaceutical firms to develop, test, and approve new prescription drugs in the U.S. is approximately 12 years (DiMasi et al., 1991). Firms spend much of this time (7–9 years) gathering the clinical evidence necessary to establish the drug's safety and efficacy (the requirements for FDA approval). After the firm collects its evidence, it submits it in the form of a new-drug application (NDA) to the FDA. The FDA then takes 1–4 years to review these applications

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and grant approval. What determines the speed at which these applications are reviewed? Does the FDA appear more responsive to some firms than to other firms? The speed of FDA approval has substantial distributional consequences for pharmaceutical firms. Hastened approval can generate tremendous profits and can give a firm some leverage over its competitors in a given market. In contrast, regulatory delays increase a firm's costs and may hinder its competitiveness. For these reasons, it is important to understand how the FDA may respond to different firms and how firm characteristics may influence the speed of FDA approval.

Previous research has examined the relationship between drug-specific characteristics and the speed of FDA review. Kaitin et al. (1991) find that the drugs rated by the FDA as therapeutically novel receive faster reviews than the drugs receiving a lower therapeutic novelty rating by the agency. In an extensive study of new molecular entities approved between 1950 and 1986, Dranove and Meltzer (1994) also find that more important drugs (using several different measures of importance) are both developed and approved more rapidly than less important drugs. Other research by Kaitin et al. (1994) illustrates how a drug's review time varies with its therapeutic category. For instance, cardiovascular drugs on average take more time to review than antiinfective drugs. While such drug-specific characteristics can explain some of the variation in review times, little attention has focused on other possible factors that may affect reviewers or review times for new drugs.

This paper empirically examines the effect of firm characteristics on the length of time required by the Food and Drug Administration (FDA) to review and approve new-drug applications between 1990 and 1992. The question of interest is whether the FDA treats all firms equally. If not, is there any systematic variation in the responsiveness of the agency to different firms? For instance, does the research intensity, reputation, or ownership of firms affect review times for drugs? This work contributes to the growing body of empirical literature that investigates bureaucratic decision making in regulatory agencies (Magat et al., 1986; Hamilton and Schroeder, 1994; Olson, 1995, 1996b). The approach taken is one that treats regulatory decisions as endogenous and explains the variation in regulatory behavior as a function of differences that exist between firms and drugs.

Results show that, controlling for drug-specific characteristics, regulators respond to firm characteristics when evaluating new drug applications. The analysis suggests that firms that invest more per sales dollar in R&D receive faster reviews from the FDA. FDA review times are also influenced by the degree to which firms specialize in the sale of pharmaceuticals. Firms that specialize in drugs receive faster reviews

than firms that are more diversified. Current experience with the FDA is also found to be valuable. Firms with more NDAs submitted to the agency receive faster reviews than firms with fewer NDAs. Finally, the drugs from foreign-owned firms receive faster reviews than the drugs from US-owned firms. In each of these cases, the firm characteristic signals information to the reviewer about either firm reputation or NDA quality. This information reduces reviewers' uncertainty about approving a dangerous or ineffective drug and leads to faster reviews.

Additional results indicate that regulators also respond to drug-specific differences. In line with the previous research, drugs that are classified as therapeutically novel by the FDA are shown to receive faster reviews than drugs receiving a standard rating. The reason is that reviewers expect more positive feedback from patient and professional groups from the approval of therapeutically novel drugs than from drugs offering no therapeutic gain. Also, review times are shown to vary across the different therapeutic drug categories. The reason is the differences in the length and complexity of NDAs in different therapeutic categories.

The analysis in this paper implies that FDA reviewers respond systematically to the differences among firms. This analysis suggests that regulatory theories of agency behavior such as the capture theory are too general to explain FDA reviewer behavior. Capture occurs when regulators favor the interests of a particular industry over consumer interests, so capture cannot explain why regulators may favor some firms over others in the same industry. The evidence in this paper suggests that regulators respond to the heterogeneity across firms in the pharmaceutical industry. The different levels of responsiveness exhibited by FDA regulators are consistent with the theory that reviewers have discretion in the regulatory process to treat different firms in different ways. Furthermore, the pattern of reviewer discretion is influenced by the information provided by firms and the signals conveyed to reviewers via firm and drug characteristics.

The paper is organized as follows. Section 2 describes a theoretical framework for understanding regulator behavior. Section 3 uses this framework to explain why review times may differ among drugs. Section 4 discusses the methodology and data used in the analysis. Sections 5 contains the results, and Section 6 has concluding remarks.

2. A Modified External-Signals Approach

The empirical framework used in this paper relies on the external-signals model introduced by Joskow (1974) and Noll (1985). The external-signals theory suggests that regulatory agencies react to the signals

that they receive or expect to receive from various groups outside the agency. The decisions made by the agency will generate both positive feedback or signals from some groups and negative feedback or signals from other groups. The model, as developed in Olson (1995, 1996b), assumes that regulators in an agency want to maximize the net positive signals that they receive from these groups. Pursuing this objective ensures that the agency possesses a base of political support among the constituencies affected by its decisions. With this support, the agency can shelter itself from changing political environments and can maintain its autonomy. Negative feedback or signals from outside groups lead to attempts by congress or other politicians to intervene in agency operations and to reduce agency autonomy.

Regulators, however, not only respond to the feedback from outside groups; they also respond to information that is provided to the agency by external sources (Magat et al., 1986). Given the informational asymmetries facing regulators and given the high cost of obtaining information, many agencies depend on information from external sources to make regulatory decisions. The information from external groups is important because it can reduce the uncertainties facing regulators and can lead to more defensible and perhaps more efficient decisions by them. Increasing the efficiency of regulatory decisions may subsequently lead to more positive feedback from outside groups such as regulated firms, consumer groups, and politicians. The external-signals model provides a convenient framework for incorporating the role of such information into the process of agency decision making. The modified external-signals model suggests that regulatory decisions will be functions of the feedback from outside groups as well as of the information supplied by external sources to the agency.

The influence of outside feedback on regulatory decisions will depend on how the agency weighs the signals from different groups. For instance, if the agency weighs the interests of the regulated industry over all other interests, then regulatory behavior corresponds to capture as described in Stigler (1971) and Peltzman (1976). If the agency weighs the interests of consumers over other interests, then regulatory behavior is consistent with the public-interest theory of regulation. If the agency weighs the feedback that it receives from Congress above all other interests, then regulatory behavior corresponds to the congressional dominance theory of Weingast and Moran (1983).[1]

1. Weingast and Moran (1983) argue that Congress has the rewards and the sanctions (i.e. budget, legislation) necessary to effectively control agency behavior. In contrast, Moe (1985) argues that agencies possess considerable autonomy in their relations with Congress, but the president through the use of political appointments and the Office of Management and Budget plays a key role in influencing agency behavior.

The responsiveness of regulators to different groups will also be influenced by the information supplied by these groups to the agency. For instance, agencies that have no independent sources of information and that must rely on the information from certain groups, such as regulated firms, may become more responsive to those groups. Among the regulated firms, regulators may become more responsive to the firms that provide the best information. In addition, regulators may infer information from the differences that exist between different firms and may use this information to reduce regulatory uncertainty and to make regulatory decisions. Because of differences in the information supplied by firms and in the information conveyed to regulators via firm characteristics, it is possible that regulators treat firms in the same regulated industry quite differently.

Although there are other internal-agency factors, such as the design of administrative procedures and agency structure (McCubbins et al., 1987, 1989) or the professional culture (Wilson, 1989, Olson, 1996a) of the agency's staff, that influence how the agency processes its information and how it responds to different groups, the analysis in this paper takes the agency's structure, process, and professional culture as given.[2,3] Specifically, this paper examines how the feedback and information from different firms may influence FDA review times for new drugs.

In a previous application of the external signals model to FDA approval decisions, Olson (1995) shows that FDA approvals for new drugs between 1971 and 1991 were responsive to the feedback from both consumers and the pharmaceutical industry. Results suggest that the agency places more weight on consumer interests in safety than on either regulated-industry interests or the preferences of current politicians. Due to the level of aggregation in the data (i.e. industry level), it was not possible to investigate the responsiveness of agency decisions to the signals or information from specific firms.

2. McCubbins et al. (1987, 1989) argue that Congress strategically chooses administrative procedures and agency structure to limit opportunities for bureaucrats to make decisions that conflict with the preferences of politicians. In addition, they argue that agency process and structure can be used to ensure that the agency remains responsive to the constituents of interest to politicians.

3. Quirk (1980) argues that the professional norms of the FDA's staff will influence the way that reviewers respond to different constituencies. Because the FDA is staffed with scientists, technicians, and physicians who are bound to the norms of the scientific and medical communities, these professional norms may conflict with the short-term political goals of congressmen or the president and may lead bureaucrats to engage in discretionary behavior. For instance, he suggests that FDA officials ignore or evade restrictions on the agency's legal authority when its decisions are perceived to have an effect on public health.

The analysis in this paper uses a more disaggregated data set to investigate FDA responsiveness to different firms in an industry. In particular, this study utilizes drug- and firm-specific data for all new molecular entities (NMEs) approved by the agency between 1990 and 1992. The agency decision variable is the length of the review time (defined as the approval date minus the application date) for each drug. With this data set it is possible to examine how review times vary by drug and to what extent reviewers respond to signals and information from firms.

3. Explaining Differences in FDA Review Times

The 1962 Drug Amendments to the Food, Drug, and Cosmetic Act of 1938 define the regulatory standard used by the FDA to evaluate NDAs. Pharmaceutical firms must show, first, that a new drug is safe, and second, that it is effective for its intended use. Furthermore, the 1962 Amendments outlined a multistage process for firms and the FDA to follow to obtain approval for a new drug. The process begins with laboratory and animal studies, and continues with three phases of clinical studies. When the firm completes all of these studies (which can take 8–10 years), it compiles all of its evidence and then submits it in the form of an NDA to the FDA for review.[4] The NDA constitutes the primary source of information reviewers use to evaluate the safety and effectiveness of each drug.

There is substantial variation in the review times for new drugs. For instance, the NDA for Videx, an antiinfective drug manufactured by Bristol-Myers Squibb, took 6.2 months to review and was approved in 1991, whereas the NDA for Daypro, an analgesic drug manufactured by Searle, took 122.5 months to review and was approved in 1992. For the sample of 74 NCEs approved by the FDA between 1990 and 1992, the mean review time was 31 months and the standard deviation was 24 months. Differences in the review times for new drugs may be due to the differences in the signals and information from firms.

The external-signals model suggests that regulators in the FDA want to maximize the positive feedback that they receive from various external groups. Regulators will utilize information about the firm and the drug in their review decisions to achieve this objective. Hence, the

4. In some cases firms may submit incomplete NDAs. An incomplete application from a firm can lengthen the review process, because regulators must first discover what additional information is needed and then reviewers must wait for the firm to submit that information.

variation in review times among different drugs will be explained by the variation in the firm- and drug-specific information or signals observed by regulators. The explanatory variables representing firm- and drug-specific information or signals fall into four categories: (1) drug characteristics that influence the FDA's perception of the feedback likely to be received from patient or professional groups after the drug's approval; (2) drug characteristics that directly affect the reviewer's uncertainty about making a Type I error (the probability of approving a dangerous or ineffective drug); (3) firm characteristics that proxy for firm reputation or NDA quality, which directly affect the reviewer's uncertainty about making a Type I error; and (4) firm characteristics that proxy for the political clout of the firm.

3.1 Drug Characteristics Affecting Expected Future Feedback

One drug characteristic that influences the reviewer's perception of the likely feedback from patient and professional groups is the therapeutic novelty of a drug. FDA reviewers want to expedite the approval of therapeutically novel drugs because the expected feedback from both patient and professional groups is expected to be greater for such drugs.

The FDA has used a rating system based on therapeutic novelty since the mid-1970s to help reviewers identify important new drugs and expedite their approval. Reviewers assign a rating to each new drug based on their best estimate of the drug's therapeutic potential. The FDA uses a drug's rating to determine its place in the queue of new drug applications to be reviewed. A drug rated type A was considered to represent an important therapeutic gain; type B represented a modest therapeutic gain; and type C represented little or no therapeutic gain. In 1987, the agency introduced type AA rating for drugs used in the treatment of AIDS. Type AA drugs were to receive expedited approval. In 1992, this rating scheme was dropped in favor of a simpler scheme which classified a drug into one of two groups: either P for priority, indicating therapeutic novelty, or S for standard, indicating little or no therapeutic gain.

In a recent study of new drugs approved between 1950 and 1986, Dranove and Meltzer (1994) show that important drugs (using several different measures of a drug's importance, including both scientific and commercial measures) are developed and approved more rapidly than less important drugs. Their results suggest that the variation in the speed of FDA review is a function of the therapeutic novelty ratings assigned to drugs by the FDA.

Variation in review times is also observed among different therapeutic categories of drugs. Within the FDA, new drugs are assigned to different departments within the Division of New Drugs for review based upon their therapeutic category. Kaitin et al. (1994) show that the mean review times for new drugs in six therapeutic categories approved between 1990 and 1992 fall into two distinct clusters. The review times for antineoplastic (17.1 mo.), antiinfective (19.6 mo.), and endocrine agents (19.9 mo.) were similar, as contrasted to those for cardiovascular (44.4 mo.), anesthetic/analgesic (46.7 mo.), and central-nervous-system agents (35.8 mo.). As described below, FDA reviewers suggest that the review times for drugs in the cardiovascular, analgesic, and central-nervous-system categories are longer than in the other categories because of the chronic nature of the illnesses. It is more difficult to establish proof of efficacy for drugs for chronic conditions than it is for drugs for acute conditions.

In addition, however, the therapeutic category of a drug may influence the reviewer's perception of the likely feedback from patient and professional groups. For example, FDA reviewers may want to expedite the approval of drugs from therapeutic categories that affect the largest numbers of people (such as cancer or antineoplastic drugs) because the expected feedback from these patient groups will exceed the feedback from smaller patient groups associated with other therapeutic drug categories.[5] If that is true, then FDA reviewers may process the applications for new drugs affecting larger patient groups more quickly than the applications for new drugs affecting smaller patient groups.

3.2 Drug Characteristics Affecting Regulator Uncertainty about Type I Error

Some therapeutic categories consist mainly of drugs for chronic conditions (e.g. cardiovascular, central nervous system, analgesics) in contrast to drugs used to treat acute illnesses (e.g. antiinfectives, endocrine, antineoplastics). The new drugs for chronic illnesses generally require longer and more complex clinical tests to evaluate long-term efficacy and toxicity (Dranove and Meltzer, 1994) than the drugs for acute ill-

5. There is also a very high demand for cardiovascular drugs, but as discussed below, there may be other factors that lead to longer review times for these drugs.

nesses.[6] This means that the applications for new drugs for chronic illnesses in these therapeutic categories will contain more information and will be more difficult for reviewers to evaluate than the NDAs for acute illnesses in other therapeutic areas. Hence, regulators face more uncertainty about the prospect of Type I error for drugs in these categories. The increased uncertainty leads reviewers to take longer to approve the drugs for chronic illnesses.

Regulator apprehension of Type I error may also be aroused by public scandals or attention focused on a particular class of drugs. For instance, Bakke et al. (1995) and DiMasi et al. (1995) suggest that safety problems involving nonsteroidal antiinflammatory drugs in the early 1980s have prompted closer monitoring of this class of drugs by the FDA. In particular, media attention generated by the the deaths associated with the drug benoxaprofen and its subsequent withdrawal from world markets in 1982 led to increased caution among FDA reviewers about other nonsteroidal antiinflammatory drugs.[7] In particular, no nonsteroidal antiinflammatory drugs were approved in the U.S. between April 1982 and December 1985, and since that time, review times for drugs in this therapeutic category (analgesics) have been longer than for drugs in other therapeutic classes.

3.3 Firm Characteristics: Reputation or NDA Quality Signals

In addition to drug characteristics, FDA reviewers may also be responsive to firm-specific characteristics. Firm-specific characteristics signal information to the reviewer about the expertise, experience, or reputation of the firm submitting the NDA. Because of the nature of the review process, the uncertainty facing reviewers, and the fear of Type I error, review times will be influenced by the reviewer's judgements and evaluations of each NDA and by the reviewer's perception and information about each firm. For instance, reviewers more familiar with the products and reputation of a particular firm may process its NDAs more quickly than those from a lesser-known firm. Repeat interactions with the scientists from a particular firm may also help expedite the review

6. For instance, cardiovascular clinical drug trials may involve as many as 10,000 patients, while clinical trials for an antineoplastic may involve only 50 patients. The differences in the trial sizes are due to the increased difficulty of showing efficacy and safety among drugs for chronic conditions and to the differences in the available populations of individuals with a particular ailment.

7. Two other compounds, zomepirac and suprofen, in this class of drugs that have been approved in the US have also been subsequently withdrawn from the market (DiMasi et al., 1995, p. 168).

process. In both cases, the information that reviewers have about firms reduces regulatory uncertainty and allows them to increase the speed of review and hence increase the positive feedback from groups that want faster reviews of drugs.

Given that firms also face uncertainty about the review process, certain firm characteristics may proxy for the degree of information that firms possess about the FDA review process. Firms with more experience in dealing with the FDA may be more familiar with the complex set of procedures used by reviewers. Experienced firms may submit better or more complete applications than inexperienced firms, so that reviewers could process the applications from these experienced firms more quickly.

Foreign firms, without strong US contingents, may be disadvantaged in the FDA approval process because of the problems in dealing with another country's regulatory laws or bureaucracy. On the other hand, most of the globally competitive pharmaceutical firms of foreign origin have well-established US subsidiaries. These multinational firms may possess advantages over less globally competitive firms in the approval process for new drugs either because of scale advantages or because they may have been able to market their drug in their home country or other countries prior to gaining US approval. This would provide the multinational firm with a broader set of evidence supporting its FDA application and lead regulators to approve the NDAs from these foreign-owned multinationals more quickly than the NDAs from less globally competitive firms.

Firm characteristics may influence a reviewer's perception of a firm's reputation, or such characteristics may be indicative of actual differences in the quality of the NDAs submitted by different firms. For instance, research-intensive firms that have specialized research departments may submit more comprehensive clinical evidence to support a new drug application than less research-intensive firms. Unfortunately, the analysis in this paper does not have access to information on NDA quality, but some of the firm-specific variables used in this analysis may well be proxies for NDA quality.

3.4 Firm Characteristics: Signals of Political Clout

Some firm characteristics may be indicators for the political clout of the firm within an industry. For instance, large firms may have more political clout than small firms because large firms have their own PACs (in addition to trade-association PACs), which lobby for the firm's own interest within the industry. Large firms also employ more individuals than small firms, and past research has shown (Olson, 1995) that regula-

tors are responsive to the employment outlook in regulated industries. Regulators may be more responsive to the interests of large firms because these firms can generate more adverse feedback for regulators than small firms. Hence, regulators may approve the NDAs from large firms faster than the NDAs from small firms.[8]

Country of firm ownership may be another indicator of political clout. US firms may have more political clout with the FDA than foreign firms because of the activity of US trade associations. However, many foreign firms have US contingents that are members of the US trade associations. In addition, there are an increasing number of PACs established by foreign firms that lobby specifically for the interests of these foreign firms within the US market. The presence of such lobbies reduces the differences in political clout between US and foreign firms. However, there may still be reputational or NDA-quality signals associated with the country of firm ownership that may cause regulatory responsiveness to differ between US and foreign firms.

3.5 Summary

Each of the drug-specific and firm-specific characteristics described above (therapeutic novelty, therapeutic category, firm reputation, firm experience with the FDA, R&D intensity, and country of firm ownership) may lead regulators to review drug applications at different speeds. Thus, some firms may face greater regulatory hurdles than other firms in the FDA approval process.

FDA personnel acknowledge that the informational asymmetries between different firms and the FDA create delays in the approval process. Since the introduction of user fees in the processing of new drug applications in 1992, the FDA has implemented new programs designed to increase the information available about the FDA review process to all firms prior to the submission of the NDA. The goal of such programs is to inform firms about the expectations of FDA reviewers regarding NDA content and quality and to prevent firms from submitting incomplete applications. Regulators at the FDA believe that such

8. Another possibility is that that large firms may produce better NDAs than small firms—NDAs that are easier for regulators to review. Henderson and Cockburn (1996) suggest that large firms have higher research productivity than small firms because the large firms realize economies of scale and scope in their research programs. These advantages may enable them to produce better NDAs than small firms. The available data makes it impossible to distinguish between scale or scope advantages and the political-clout hypothesis for explaining why large firms may receive faster reviews than small firms.

programs will help reduce regulatory delay in the approval process for new drugs.

The next section describes the methodology and data used to examine the relationship between FDA review times and the drug-specific and firm-specific information or signals.

4. Methodology and Data

The regulatory decision variable r_{it}, FDA review times, is modeled as a function of the drug-specific characteristics X_{it} and firm-specific characteristics Y_{it}:

$$r_{it} = \alpha + \delta X_{it} + \beta Y_{it} + \epsilon_{it}.$$

The coefficients δ and β measure the extent to which regulatory review times for new drugs are influenced by drug-specific and firm-specific characteristics, respectively. The vector of drug-specific characteristics X_{it} consists of seven dummy variables. The first of these is a dummy variable *priority* indicating whether the new drug has received a priority rating from the FDA. This variable is equal to 1 for all new drugs classified as P for priority in 1992 or as a type A, AA, or B in preceding years. The variable takes on a value of 0 for all new drugs classified as S for standard in 1992 or as C in the preceding years. An alternative specification will examine a different classification of priority drugs in which the dummy variable *priority*2 is set equal to 1 for all new drugs classified as P in 1992 or as A or AA in preceding years, and 0 for all new drugs classified as S in 1992 or as B or C in preceding years. Since the FDA developed the rating system to help expedite the review of important drugs, the coefficient for this variable is predicted to be negative, indicating that priority drugs have shorter review times.

The remaining variables are a series of six dummy variables representing the different therapeutic categories associated with the drugs. These are *cardio* for cardiovascular drugs, *neopl* for antineoplastic drugs, *infect* for antiinfective drugs, *endo* for endocrine drugs, *cns* for central-nervous-system drugs, and *analges* for analgesic/anesthetic drugs. These variables are included to control for differences in the review times among drugs from the various therapeutic categories. The signs for *cardio*, *cns*, and *analges* are predicted to be positive because many of these drugs are used to treat chronic illnesses and the corresponding NDAs are expected to be more complex and difficult to review than the NDAs in other therapeutic categories. In addition, the sign for *an-*

alges may be positive because of the increased regulatory concern about the safety of nonsteroidal antiinflammatory drugs and the adverse effects associated with taking these drugs. Partitioning the data differently (in two general categories for chronic and acute) does not alter the substantive results or increase the explanatory power of the model.

The vector of firm-specific characteristics includes the following variables. The first variable *rd/tsale* is a variable measuring the research intensity of the firm. It is constructed as the ratio of aggregate R&D to aggregate sales for a firm. Firms that are more research-intensive should receive faster reviews from the FDA than firms that are less research-intensive. Firms that invest a higher proportion of their sales revenue in R&D show a commitment to research. The firm's reputation with regulators may be favorably influenced by its commitment to R&D. In addition, research-intensive firms may submit more comprehensive clinical evidence in support of an NDA or may respond more quickly to regulators' requests for more information. Hence, the predicted sign for this variable is negative. In contrast to R&D intensity, an alternative specification will examine if aggregate R&D expenditures $\ln(R\&D)$ also influence NDA review times.

The second variable, *dsale/tsale*, is a measure of firm diversification. It is constructed as the ratio of a firm's total pharmaceutical sales to its total sales. Reputation effects may cause regulators to view less diversified pharmaceutical firms as specialists or experts in the production and sale of pharmaceuticals. Because these firms are viewed as specialists, they may receive faster regulatory reviews than firms that are more diversified. In addition, firms that specialize in the production of pharmaceuticals may have more familiarity with FDA regulations or FDA regulators. The increased information that these firms possess about the FDA approval process may enable them to produce higher-quality NDAs and hence receive faster reviews. This corresponds to a predicted negative coefficient for *dsale/tsale* variable.

The third variable, *applic*, is a measure of a firm's current experience in dealing with the FDA. The variable is the number of a firm's NDAs in this sample that are submitted to the FDA (not yet approved) at the time of drug *i*'s approval. Firms with a greater number of NDAs submitted to the agency may have more contact with regulators in the agency or may have more familiarity with current regulatory rules and norms. Also, regulators may be more familiar with the scientists from firms that have several NDAs pending. For these reasons, the predicted sign of the coefficient for *applic* is negative.

To distinguish between current experience and past experience

with the agency, an alternative specification will include the variable *aprl*, which is the cumulative number of the firm's new drugs approved by the agency since 1990 prior to the approval of drug *i*. Since prior approvals may reflect learning on the part of the firm about the regulatory process and may increase regulators' familiarity with a particular firm, the predicted sign of the coefficient for *aprl* is negative. In addition, a comparison of the coefficients for *applic* and *aprl* in the regression will indicate whether a firm's current experience or past experience will have a greater influence on the review time for an NDA.

The fourth variable, *usfirm*, is a dummy variable equal to 1 if the firm submitting the new drug application is a US firm and equal to zero otherwise. The coefficient for this variable indicates whether the drugs from US firms receive faster or slower FDA reviews than the drugs submitted by foreign firms. That might happen because of the difficulties that foreign firms may encounter in dealing with the US bureaucracy. This would correspond to a negative coefficient for *usfirm*. However, the drugs from foreign firms may receive faster FDA reviews than the drugs from US firms because foreign firms submit higher-quality NDAs. Reasons for higher application quality from foreign firms may be differences in the design of animal or clinical tests, differences in the number of patients used in the clinical tests, or differences in the amount of material and evidence contained in the application. This would correspond to a positive coefficient for *usfirm*. Hence, the predicted sign for *usfirm* is ambiguous.

An alternative specification will include the variable ln(*employ*), which is the logarithm of the total number of employees of a firm at time $t - 1$. This variable is included to determine if the applications variable is a proxy for firm size. Larger firms may have some economies of scale in dealing with FDA regulators. Also, firm size may be a proxy for political clout with regulators. For these reasons, larger firms may receive faster reviews for their drugs than smaller firms. This corresponds to a predicted negative coefficient for ln(*employ*).

In addition, another specification will examine if there are interaction effects between firm- and drug-specific characteristics. Specifically, the analysis will examine the interaction of therapeutic novelty with each of the firm-specific variables to determine if there is more of a signaling role for firm-specific information for therapeutically novel drugs. Regulators may be more responsive to certain firm characteristics when the NDA constitutes a significant therapeutic gain.

A linear regression is estimated using panel data representing the set of new chemical entities approved between 1990 and 1992. A specification for the regression is given below, and the predicted signs for the coefficients that are not ambiguous are listed in parentheses:

$$
\begin{aligned}
r_{i,t} = \alpha_t &+ \delta_1^{(-)} priority_{i,t} + \delta_2 cardio + \delta_3 neopl \\
&+ \delta_4 infect + \delta_5 cns + \delta_6 analges + \delta_7 endo \\
&+ \beta_1^{(-)} \frac{rd_{i,t-1}}{tsale_{i,t-1}} + \beta_2^{(-)} \frac{dsale_{i,t-1}}{tsale_{i,t-1}} \\
&+ \beta_3^{(-)} applic_{i,t} + \beta_4^{(-)} usfirm_{i,t} + \epsilon_{it}. \qquad (1)
\end{aligned}
$$

There are three econometric issues to address. First, because panel data are used, fixed-time effects are included in (1). This takes the form of separate intercepts for each of the years 1990–1992 in the sample. Given that there are 72 observations and 35 firms represented in the sample, including fixed-firm effects would severely limit the degrees of freedom. For this reason, dummy variables representing fixed-firm effects are omitted. Instead, the analysis includes the firm-specific variables described above to control for the systematic differences between firms.

Second, it is important to discuss the possible endogeneity of some of the regressors. Although current FDA review times may influence firms' incentives to submit other NDAs, I treat the applications variable *applic* as exogenous because it takes pharmaceutical firms many years to compile the evidence needed to submit an NDA to the FDA. Since current firm R&D and current firm sales may also be influenced by current FDA review times, I use a one-year lagged value for *rd*, *tsale*, and *dsale* in (1). Data limitations prevent me from using longer lags for these variables. However, reviewers are more likely to process current or recent signals from these firms than older signals. Third, because the error terms ϵ_{it} are heteroskedastic, White standard errors are used for all hypothesis tests.

This analysis focuses on the review times for new chemical entities (NCEs) approved by the FDA. Kaitin et al. (1994) define an NCE ''as any new molecular compound not previously approved in the United States, excluding biologics, vaccines, and diagnostic agents. New salts, esters, and dosage forms of previously approved compounds are also excluded.'' A total of 79 new drugs were approved by the agency between 1990 and 1992. Of these, 74 met the definition of an NCE. Two NCEs were dropped from my sample because of a lack of firm-specific data, so that the regression in (1) is performed on a sample of 72 NCEs.

The data used in this analysis come from the following sources. The new chemical entities approved by the FDA, the review times, the approval dates, and the therapeutic ratings assigned by the FDA are all taken from Kaitin et al. (1994). The therapeutic categories assigned to these drugs were also provided by Kaitin et al. Each firm's research

Table I
Means and Standard Deviations for Variables over 1990–1992

Symbol	Definition	Mean	SD	Min	Max
r	Review time (months)	31	24	1.3	122.5
rd/tsale	R&D/total sales	0.12	0.14	0.02	1.2
dsale/tsale	Drug sales/total sales	0.51	0.26	0	1
applic	Other FDA applications	2.2	1.6	1	8
usfirm	Dummy for US firm	0.61	0.49	0	1
employ	Firm employment	44,686	27,135	175	172,890
aprl	Prior drug approvals	1.06	1.5	0	7
rd	R&D (million 1990 dollars)	615	333	3	1,663
priority	Dummy for priority drugs	0.47	0.50	0	1
*priority*2	Highest priority drugs	0.32	0.47	0	1
cardio	Cardiovascular dummy	0.24	0.43	0	1
neopl	Neoplastic dummy	0.10	0.30	0	1
infect	Antiinfective dummy	0.26	0.44	0	1
cns	Central nervous system dummy	0.11	0.32	0	1
analges	Anesthetic/analgesia dummy	0.17	0.38	0	1
endo	Endocrine dummy	0.10	0.30	0	1

and development expenditures, pharmaceutical sales, total sales, employment, and firm ownership information was obtained from the annual reports of the firms for various years. Summary statistics for the variables used in this analysis are reported in Table I.

5. Results

For the sample of 72 NCEs used in this analysis the mean review time is 31 months with a standard deviation of 24 months. This suggests that there is quite a bit of variance among the review times for the different NCEs. Within this sample, 34 drugs received a priority rating (type A, AA, B, or P), while the remaining 38 drugs received a standard rating (type C or S). The mean review time for the class of priority

NCEs was 20 months, while the mean review time for the class of standard NCEs was 40 months.

A total of 35 firms are represented in this sample: 19 US firms and 16 foreign firms. The US firms submitted a total of 44 NCE applications to the FDA during the period, 19 of which received a priority rating. Foreign firms submitted a total of 28 NCE applications, 15 of which received a priority rating.

Surprisingly, the average review times for the drugs from foreign firms were smaller than for the drugs from US firms. The average review time for the 44 NDAs submitted by US firms was 35.3 months, while that for the 28 NDAs submitted by foreign firms was 23.5 months. This difference was also seen within the classes of priority and standard drugs from US and foreign firms. Among the priority drugs, the average review time for drugs from US firms is 23.6 months while the average review time for drugs from foreign firms is 15.5 months. Among the drugs receiving a standard rating from the FDA, the average review time for drugs from US firms is 44.2 months, while the average review time for drugs from foreign firms is 32.6 months.[9]

Of the 35 US and foreign firms, 19 firms submitted more than one application; 9 firms submitted more than two applications; 5 firms submitted more than three applications. The largest number of applications submitted by a single firm was eight by Bristol-Myers Squibb.[10]

Not surprisingly, most of the firms in the sample represent large pharmaceutical firms or their subsidiaries. The average employment among these firms is 44,686. On average, a firm's pharmaceutical sales represent over 50% of its total sales. Only one firm in the sample had no pharmaceutical sales prior to the approval of its drug; this firm did have sales in other areas besides pharmaceuticals. Firms in this sample are research-intensive, investing an average of 12% of total sales in R&D, with a standard deviation of 14%.

The regression results for (1) are presented in the first column of Table II. Results indicate that both drug-specific characteristics and firm-specific characteristics are important determinants for the length of regulatory review times for drugs. The intercept terms, which are not reported for each of the different years, were also highly significant.

9. The pattern of shorter review times for the drugs from foreign firms is seen in four of the six therapeutic categories: analgesic, central nervous system, antiinfective, and cardiovascular drugs.

10. Only three applications in the sample were submitted jointly by two firms. Two alliances were between a US chemical firm and a US pharmaceutical firm. The other alliance was between a US and a foreign firm. In each case, data from the firm considered the primary maker of the drug was used in the analysis.

Table II
Regression Results[a]: Dependent Variable = FDA Review Time

Variable	(1)	(2)
rd/tsale	−25.30	−50.67
se	(10.10)***	(18.5)***
dsale/tsale	−18.45	−23.46
se	(7.64)***	(8.80)***
applic	−6.03	−5.75
se	(1.71)***	(1.69)***
usfirm	19.23	18.32
se	(5.11)$^{+++}$	(5.45)$^{+++}$
ln(*employ*)	—	−4.50
se	—	(2.94)*
aprl	—	0.65
se	—	(1.35)
priority	−5.56	−4.63
se	(3.87)*	(4.00)
cardio	18.45	19.37
se	(7.66)$^{++}$	(7.50)$^{++}$
cns	20.55	20.63
se	(11.77)$^{+}$	(11.30)$^{+}$
analges	30.53	30.80
se	(12.56)$^{++}$	(12.49)$^{++}$
neopl	−11.66	−14.69
se	(8.58)	(8.36)$^{+}$
infect	−4.17	−4.20
se	(6.72)	(6.24)
endo	−4.88	−6.07
se	(7.67)	(7.42)
Fixed time effects	Significant	Significant
N	72	72
Adj. R^2	0.77	0.77

[a] Standard errors (se) are presented in parentheses. Significance at the .1, .05, and .01 levels is indicated by *, **, *** for a one-tailed test and by $^{+}$, $^{++}$, $^{+++}$ for a two-tailed test.

The intercepts indicate a pattern of declining average review times over the period. The model explains approximately 77% of the variation in FDA review times.

As predicted, firms investing a higher proportion of their sales in R&D receive shorter review times than firms that have lower R&D-to-sales ratios. The coefficient for *rd/tsale*, which is negative and significant at the .01 level, indicates that a 10% increase (almost 1 standard deviation) in the lagged R&D-to-sales ratio reduces the review time for

a drug by 2.53 months. The result suggests that a firm's research intensity is a characteristic that signals information to the reviewer about firm reputation or NDA quality. Such information reduces the reviewer's uncertainty about making a Type I error and leads to faster reviews for NDAs from research-intensive firms.

Firms that specialize in pharmaceutical sales also have shorter review times than more diversified firms. The coefficient for the diversification variable *dsale/tsale*, which is negative and significant at the .01 level, shows that a 10% increase in the lagged ratio of pharmaceutical sales to total firm sales reduces the review time for a drug by 1.84 months. The diversification variable also signals information to the reviewer about firm reputation or NDA quality. This result suggests that there are reputational advantages or other informational advantages that accrue to specialized pharmaceutical firms in the regulatory process.

Current experience in dealing with the FDA also leads to shorter review times for new drugs. The coefficient for *applic*, which is negative and significant at the .01 level, suggests that a one-application increase in the number of other NDAs submitted to the FDA reduces the review time for a drug by 6 months. A firm with multiple applications submitted to the agency may have more contact with FDA regulators. Such contact could lead to better information channels between the scientists at the firm and regulators, or it could produce additional reputational advantages for the firm among FDA reviewers. The result suggests that current experience is another firm characteristic that signals information to reviewers. Such information reduces regulatory uncertainty and leads to faster reviews for new drugs.

The dummy variable for a US firm is significantly positive at the .01 level. The coefficient shows that a US firm experiences a regulatory review for its drug that is 19 months longer than a review time for a foreign firm. This suggests that US firms are not advantaged in the FDA regulatory process. Foreign firms are receiving faster reviews than US firms. The result holds when one controls for the drug's therapeutic novelty and the firm-specific characteristics. Given these controls, the result suggests that foreign firms either have reputational advantages or submit higher-quality applications than US firms.

If a foreign firm is able to market its drugs in its home country prior to gaining FDA approval, then it may be able to compile a more convincing or conclusive set of evidence in support of the drug's US application. Also, regulators may face less uncertainty about any adverse consequences (e.g. deaths) associated with the drug if it is being used in another country. The reduction in regulatory uncertainty could contribute to the faster reviews for the drugs from foreign firms. The

problem with the prior-approval explanation is that many of the drugs in the sample, including many of the drugs from US firms, have been approved for use in other countries prior to US FDA approval.[11] Furthermore, there does not appear to be a direct correlation between prior approval in a foreign country and quick approval in the US. It is difficult to pinpoint the reason why foreign firms may submit higher-quality applications than US firms without having more detailed information about application quality.

Another alternative explanation relates to the commercial importance of the drug. Dranove and Meltzer (1994) showed that commercially important drugs are developed and approved more rapidly in the US than commercially unimportant drugs. Because of the cost of gaining FDA approval, foreign firms may only attempt to market the drugs in the US that they expect to be commercially successful. However, subsequent tests, which include a proxy for the commercial importance of a drug, did not provide support for this argument.[12]

Consistent with the results of Kaitin et al. (1991) and Dranove and Meltzer (1994), priority drugs are shown to receive faster reviews. The coefficient for *priority*, which is negative and significant at the .10 level, shows that the review time for a drug rated by the agency as therapeutically novel is 5.6 months shorter than for a drug receiving a standard rating from the agency. Therapeutic novelty is a drug characteristic that influences the reviewer's perception of the likely feedback from patient and professional groups. Reviewers approve therapeutically novel drugs faster than other drugs because the expected feedback associated with novel drugs is greater than that associated with standard drugs.

In addition, the coefficients for cardiovascular drugs, central-nervous-system drugs, and anesthetic/analgesic drugs are all significantly positive at the .05, .10, and .05 levels, respectively, which suggests that

11. Information contained in *de Haen's Drug Product Index International*, v. II, 1989, shows that 27 of the NCEs in this sample were approved in foreign markets prior to 1989 or prior to their approval in the US market. Among the 27, 18 were submitted to the FDA by US firms.

12. To examine if *usfirm* may be capturing some selection bias on the part of foreign firms regarding the drugs they choose to introduce into the US market, I collected information about 1995 sales performance of the drugs in the sample. Given the difficulties of collecting individual-drug sales data, I relied on the ''Top 100 Pharmacy Products by Dollar Volume'' reported in *Drugstore News for Pharmacists* for 1995. The variable *top*100 is a dummy variable set equal to 1 if the drug was on this Top 100 list in 1995; otherwise it is equal to zero. Eighteen of the drugs in my sample were on the Top 100 list. The analysis showed that when the variable *top*100 was included in (1), its coefficient was not significantly different than zero. Also, its inclusion did not alter the significance or magnitude of the coefficient for the *usfirm* dummy.

the drugs in these therapeutic categories take longer to review than drugs in other therapeutic categories. The coefficients for antineoplastic drugs, antiinfective drugs, and endocrine drugs, although negative, are not significantly different from zero. These results suggest that some therapeutic categories (cardiovascular, central nervous system, and anesthetic/analgesic) represent drug characteristics that increase the regulator's uncertainty about making a Type I error. The reason is that drugs in these categories are generally used to treat chronic conditions and the NDAs are longer, more complex, and more difficult to evaluate for them than in other therapeutic categories. In addition, scandals associated with the use of analgesics may have influenced reviewer uncertainty about drugs in this category.

Most of the results discussed above, particularly for the firm-specific variables, are robust across the specifications in Table II. The second column of Table II presents the results from the regression containing the two additional variables reflecting firm size and prior drug approvals. The coefficient for the size variable, ln(*employ*), is negative and significant at the .1 level, indicating that the NDAs from large firms receive faster reviews than those from smaller firms. The result suggests that large firms may possess some advantages over small firms in the NDA review process. These differences may be due either to scale advantages or to the political clout of the larger firms. Furthermore, the inclusion of the size variable in (2) does not greatly diminish the magnitude or the significance of the applications variable, which suggests that *applic* is not simply a proxy for firm size.[13] Instead, the firm-size variable signals additional information to reviewers about reputation or political clout of the firm.

The insignificance of the coefficient for *aprl* in (2), combined with the significance of *applic*, suggests that current experience with the FDA, not past experience, leads to faster reviews for new drugs.[14] Hence, a firm's current experience signals more information to reviewers than its past experience with the FDA.

Three alternative specifications were also estimated. In the first, aggregate R&D expenditures, ln(*R&D*), were included in (1) instead of research intensity. Although the coefficient for ln(*R&D*) was positive, it was not significantly different than zero.[15] This result suggests that a firm's aggregate R&D on all of its products is a more noisy and perhaps

13. The simple correlation between ln(*employ*) and the *applic* variables is .17.

14. The correlation between *applic* and *aprl* variables is −.07.

15. The coefficient for ln(*R&D*) was 1.36, and its standard error was 2.03. Other results were unaffected by this substitution.

less informative signal than the firm's research intensity. Hence, reviewers are less responsive to aggregate R&D expenditures than to the research intensity of the firms.

In the second specification, *priority*2 was substituted for *priority*. The variable *priority*2 includes only those drugs classified as either an important therapeutic gain (type A or P) or an AIDS drug (type AA). Although the coefficient for *priority*2 was negative, it was not significantly different than zero.[16] The only difference between this variable and *priority* is in the classification of drugs offering a modest therapeutic gain (type B). Including type B drugs among the set of standard drugs introduced more noise into the *priority*2 measure. The lack of significance of *priority*2 does not contradict the finding that regulators approved drugs offering both high and modest therapeutic gains faster than drugs that offered no therapeutic gain. It merely implies that drugs rated as the most therapeutically novel did not receive significantly faster reviews than drugs offering a modest or no therapeutic gain.

A third specification examined if there were interaction effects between therapeutic novelty (*priority*) and four firm-specific characteristics (*usfirm*, *rd/tsale*, *dsale/tsale*, *applic*). The interaction terms were included to determine if there is more information signaling of firm characteristics for therapeutically novel drugs than for standard drugs. Of the four interaction variables that were introduced into (1), only the coefficient for one, *priority·usfirm*, was significantly different from zero. In contrast to the positive coefficient for *usfirm*, the coefficient for the interaction variable *priority·usfirm* was negative, indicating that the country of a firm's ownership conveys different information for therapeutically novel drugs than for standard drugs. The coefficient indicates that novel drugs from US firms are less disadvantaged in the review process than standard drugs from US firms. However, the net effect of the *usfirm* coefficient and the interaction-variable coefficient is still positive.[17] The lack of significance of three of the four interaction variables suggests that the signaling role of information about the firm (R&D intensity, diversification, experience) does not vary with the therapeutic novelty of the drug. Drug and firm signals appear to be distinct.

16. The coefficient for *priority*2 was -3.43, and its standard error was 3.31. Other results were unaffected by the inclusion of this variable and so are not reported.

17. The net effect is given by the sum of the coefficients for *usfirm*, *priority*, and *usfirm·priority*, respectively ($26 + 0 - 14 = 12$ months). The coefficient for the *priority* variable became insignificant when the interaction variable was included in (1), and so its coefficient is set equal to zero.

6. Conclusions

Reviewers of NDAs in the FDA respond systematically to the differences that exist between firms and between drugs. The reason is that the firm and drug characteristics provide information to the reviewer. Some of this information provides a signal of firm reputation or NDA quality, which reduces the reviewer's uncertainty about making a Type I error and leads to faster reviews of the NDAs. Research intensity, specialization in pharmaceuticals, current experience with the FDA, and foreign ownership are four firm characteristics that signal this type of information to regulators. Differences in the therapeutic category of a drug signal differences in the length and complexity of the NDA.

Other characteristics influence the FDA's perception of the likely feedback that it will receive from patient and professional groups. For instance, FDA reviewers expedite the review of therapeutically novel drugs because the expected feedback from patient and professional groups is greater for drugs offering therapeutic gains than for drugs offering no therapeutic gains. Also, drugs from larger firms receive faster reviews than drugs from smaller firms, perhaps because larger firms have more political clout with the agency than smaller firms.

For new-drug approvals, the pattern of variation in review times suggests that reviewers have discretion in the review process. The presence of such discretion means that factors such as the information available to reviewers or reviewers' perceptions about firms influences the length of the review period. In addition, the pattern of reviewer responsiveness has implications for firm strategies in dealing with the FDA. In particular, the results suggest that there may be opportunities for some firms to hasten the review of their drugs by forming alliances with other firms that either are larger, are more research-intensive, are more specialized, or have more current experience with FDA reviewers. If such alliances are not possible or too costly, it may be advantageous for some firms, unfamiliar with either clinical test design strategy or the complicated FDA approval procedures, to hire experts to help them design tests, prepare the NDA, and communicate with the FDA reviewers. Such strategies will provide benefits to firms and reviewers in the review process.

The analysis also suggests that regulatory theories of agency behavior such as capture are too general to characterize regulatory decision-making in this agency. Capture theories cannot explain the heterogeneity of regulatory responses to firms in a specific industry. Capture cannot explain why regulators are more responsive to some firms than to others in the same industry. Instead, differences in the information provided by firms and differences in firms' ability to signal reputation,

expertise, and political clout appear to be important factors explaining regulatory decision making. In conclusion, this study suggests that researchers should focus more on the micro-level decisions made by regulators and the information and trade-offs facing regulators at the micro level to find explanations for the variation observed in regulatory decisions. Then, with such information, researchers can begin to build and test more micro-based theories of regulatory behavior.

References

Bakke, O.M., M. Manocchia, F. de Abajo, K. Kaitin, and L. Lasagna, 1995, ''Drug Safety Discontinuations in the U.K., the U.S. and Spain from 1974–1993: A Regulatory Perspective,'' *Clinical Pharmacology & Therapeutics*, 58(1), 108–117.

DiMasi, J.A., R. Hansen, H. Grabowski, and L. Lasagna, 1991, ''The Cost of Innovation in the Pharmaceutical Industry,'' *The Journal of Health Economics*, 10, 107–142.

DiMasi, J.A., R. Hansen, H. Grabowski, and L. Lasagna, 1995, ''Research and Development Costs for New Drugs by Therapeutic Category,'' PharmacoEconomics, 7(2), 152–169.

Dranove, D. and D. Meltzer, 1994, ''Do Important Drugs Reach the Market Sooner?'' *Rand Journal of Economics*, 25, 402–423.

Hamilton, J. and C. Schroeder, 1994, ''Strategic Regulators and the Choice of Rulemaking Procedures: The Selection of Formal vs. Informal Rules in Regulating Hazardous Wastes,'' *Law and Contemporary Problems*, 57, 111–160.

Henderson, R. and I. Cockburn, 1996, ''Scale, Scope, and Spillovers: The Determinants of Research Productivity in Drug Discovery,'' *Rand Journal of Economics*, 27(1), 32–59.

Joskow, P., 1974, ''Inflation and Environmental Concern: Structural Change in the Process of Public Utility Price Regulation,'' *Journal of Law and Economics*, 17, 291–327.

Kaitin, K.I., N.R. Phelan, D. Raiford, and B. Morris, 1991, ''Therapeutic Ratings and End-of-Phase II Conferences: Initiatives to Accelerate the Availability of Important New Drugs,'' *Journal of Clinical Pharmacology*, 31, 17–24.

Kaitin, K.I., M. Manocchia, M. Seibring, and L. Lasagna, 1994, ''The New Drug Approvals of 1990, 1991, and 1992: Trends in Drug Development,'' *The Journal of Clinical Pharmacology*, 34, 120–127.

Magat, W., A. Krupnick, and W. Harrington, 1986, *Rules in the Making: A Statistical Analysis of Regulatory Agency Behavior*, Washington, DC: Resources for the Future.

McCubbins, M., R. Noll, and B. Weingast, 1987, ''Administrative Procedures as Instruments of Political Control,'' *Journal of Law, Economics, & Organization*, 3, 243–277.

——, 1989, ''Structure and Process, Politics and Policy: Administrative Arrangements and the Political Control of Agencies,'' *Virginia Law Review*, 75, 431–482.

Moe, T., 1985, ''Control and Feedback in Economic Regulation: The Case of the NLRB,'' *American Journal of Political Science*, 79, 1094–1117.

Noll, R., 1985, ''Government Regulatory Behavior: A Multidisciplinary Survey and Synthesis,'' in R. Noll, ed., *Regulatory Policy and the Social Sciences*, Berkeley: University of California Press.

Olson, M., 1995, ''Regulatory Agency Discretion among Competing Industries: Inside the FDA,'' *Journal of Law, Economics, & Organization*, 11(2), 379–405.

——, 1996a, ''Explaining Regulatory Behavior in the FDA: Political Control vs. Agency Discretion,'' in G. Libecap, ed., *Advances in the Study of Entrepreneurship, Innovation, and Economic Growth*, v. 7, JAI Press Series.

——, 1996b, ''Substitution in Regulatory Agencies: FDA Enforcement Alternatives,'' *Journal of Law, Economics, & Organization*, 12:2, 376–407.

Peltzman, S., 1976, ''Toward a More General Theory of Regulation,'' *Journal of Law and Economics*, 19, 211–240.

Quirk, P.J., 1980, ''The Food and Drug Administration,'' in J.Q. Wilson, ed., *The Politics of Regulation*, Basic Books.

Stigler, G., 1971, ''The Theory of Economic Regulation,'' *Bell Journal of Economic and Management Science*, 2, 2–21.

Weingast, B. and M. Moran, 1983, ''Bureaucratic Discretion or Congressional Control: Regulatory Policymaking by the Federal Trade Commission,'' *Journal of Political Economy*, 91, 765–800.

Wilson, J.Q., 1989, *Bureaucracy: What Government Agencies Do and Why They Do It*, Basic Books.

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